1. Project Planning & Management

Initiative: Digital Egypt Pioneers Initiative (DEPI) – Round 5
Track: Microsoft Data Engineer
Company: Skills Dynamix
Group Code: ALX5_AIS5_S5
Project Name: Egyptian Ministry of Interior (MOI) Crime News Intelligence Pipeline
Supervisor / Lecturer: Eng. Ahmed Azab
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Team Members

Member Name	Project Role	Email Address
Ahmed Ali Ahmed Ali	Team Leader	ahmed.a.ali.tech@gmail.com
Omar Tamer Mostafa Elhadidi	Data Engineer	omar.t.elhadidi@gmail.com
Mahmoud Ibrahim Abdul Qader	Data Engineer	mahmoud.ibrahim.eg1@gmail.com
Sondos Ahmed Mabrouk Habiba	Data Engineer	sondosahmed5425@gmail.com
Konoz Nashaat	Data Engineer	konoznashaat123@gmail.com
Shahd Nezar Atef Hagag	Data Engineer	shahdhagag.ss@gmail.com

Project Proposal

Overview of the Project:
The Egyptian Ministry of Interior (MOI) Crime News Intelligence Pipeline is an automated data engineering platform. The system ingests over 4,700 historical and daily public security reports published on the official Egyptian Ministry of Interior Facebook page. Raw unstructured Arabic statements are collected, normalized, and parsed to extract key incident features. The data moves through a standard Medallion architecture (Bronze JSON → Silver Parquet → Gold PostgreSQL Star Schema), automated by Apache Airflow, and feeds an interactive analytics dashboard with geographical heatmaps across Egyptian governorates.

Objectives:

1. **Automated Ingestion:** Periodically extract official crime reports, capturing post ID, publication timestamp, Arabic news text, and media links.
2. **Arabic Text Processing:** Clean narrative Arabic text, remove special formatting noise, and extract core incident attributes (incident date, governorate, crime category, legal action).
3. **Dimensional Data Warehouse:** Load clean data into a Star Schema data warehouse in PostgreSQL with full referential integrity.
4. **Orchestration & Reliability:** Schedule and manage the pipeline using Apache Airflow DAGs with automatic error retries and alerting.
5. **Data Quality Assurance:** Run automated data invariant tests using pytest to prevent schema drift, null primary keys, and orphan foreign keys.
6. **Business Intelligence Serving:** Provide an interactive Streamlit dashboard featuring incident KPIs, monthly crime distributions, and density heatmaps across Egypt's 27 governorates.

Scope Boundaries:

- **In-Scope Capabilities:** Scraping historical posts (4,700+ articles) and incremental batches; Medallion architecture (Bronze JSON, Silver Parquet, Gold PostgreSQL); idempotent loading (`ON CONFLICT DO NOTHING`); automated pytest invariants; daily Airflow scheduling; and Streamlit analytics visualization.
- **Out-of-Scope Boundaries:** Access to private or internal police department databases; real-time audio/video surveillance stream analysis; and automated sentencing or predictive policing decisions.

Project Plan

Timeline (Gantt Chart / Roadmap):

Phase & Target Dates	**Milestone Deliverable**	**Technical Focus Areas**
Phase 1 (Oct 2 – Oct 16)	Project Proposal & Requirements	Team formation, project proposal, requirements gathering, repository setup.
Phase 2 (Oct 17 – Oct 30)	Bronze & Silver Engineering	Raw post scraping (`extract_crime_news.py`), Arabic text cleaning, Parquet transformation (`transform_silver.py`).
Phase 3 (Oct 31 – Nov 6)	Star Schema Warehouse (Gold)	PostgreSQL Star Schema DDL (`gold_schema.sql`), batch loader (`load_gold_warehouse.py`), indexing.

Phase & Target Dates	Milestone Deliverable	Technical Focus Areas
Phase 4 (Nov 7 – Nov 20)	Orchestration & Automated QA	Airflow DAG creation (`dags/moi_crime_dag.py`), retries configuration, pytest quality gates (`test_warehouse.py`).
Phase 5 (Nov 21 – Nov 30)	Dashboard & Visualization	Streamlit app (`app.py`), Folium/Plotly Egypt heatmaps, filtering by governorate/date, cloud deployment.
Phase 6 (Dec 1 – Dec 4)	Final Verification & Delivery	End-to-end integration testing, technical documentation, user manual, final project defense.

Milestones & Deliverables:

- **Milestone 1 (Oct 16):** Finalized documentation, schema specifications, and git repository (`NHA-5-079`).
- **Milestone 2:** Automated ingestion script saving raw posts to `data/bronze_crime_news.json`.
- **Milestone 3:** Clean Silver dataset of 4,700+ records in `data/silver_crime_news.parquet`.
- **Milestone 4:** Star Schema DDL (`sql/gold_schema.sql`) and warehouse loader (`src/load_gold_warehouse.py`).
- **Milestone 5:** Automated test suite (`tests/test_warehouse.py`) passing 5 core invariants.
- **Milestone 6:** Airflow DAG (`dags/moi_crime_dag.py`) running daily scheduled orchestration.
- **Milestone 7:** Live Streamlit application (`app.py`) displaying KPIs and interactive heatmaps.

Resource Allocation:

- **Software & Libraries:** Python 3.11, Pandas, PyArrow, SQLAlchemy, Psycopg2-binary, Pytest, Streamlit, Plotly.
- **Database & Compute:** PostgreSQL 16 warehouse, Parquet storage, Apache Airflow 2.8+ on GitHub Codespaces.
- **Infrastructure Cost:** 100% free and open-source tools with zero cloud hosting bills.

Task Assignment & Roles

Team Member	Project Role	Defined Responsibilities & Deliverables
Ahmed Ali Ahmed Ali	Team Leader	Overall project coordination, data scraping supervision, and GenAI / LLM entity extraction benchmarking.
Omar Tamer Mostafa Elhadidi	Data Platform Engineer	End-to-end pipeline architecture, Star Schema dimensional modeling, batch loading optimization, Airflow DAG orchestration, and automated pytest quality suites.
Mahmoud Ibrahim Abdul Qader	Data Modeling Engineer	Entity-Relationship (ER) diagram design, SQL DDL table creation (`DIM_DATE`, `DIM_LOCATION`, `DIM_CRIME_TYPE`, `FACT_CRIME_NEWS`), and index tuning.
Sondos Ahmed Mabrouk Habiba	BI & Analytics Engineer	Streamlit frontend design, interactive filtering widgets, monthly trend charts, and Egypt density map visualization.
Konoz Nashaat	Quality Assurance Engineer	Data validation rules, edge-case testing (missing dates, unrecognized governorates), and pipeline integration test execution.
Shahd Nezar Atef Hagag	Requirements & Documentation	Requirements documentation, stakeholder interview notes, system analysis reports, user manual, and final presentation slide deck.

Risk Assessment & Mitigation Plan

Identified Risk & Severity	Technical Mitigation Solution
Unstructured Arabic text variations (High)	Build modular cleaning functions in `src/transform_silver.py` with lookup dictionaries for Egyptian governorates and incident keywords.
Arabic date format inconsistencies (High)	Implement multi-pattern date parsing handling both Hijri/Gregorian dates, Arabic month names (e.g., أكتوبر, نوفمبر), and standard numeric formats.

Identified Risk & Severity	Technical Mitigation Solution
LLM latency and API rate limits (High)	Rely on optimized regex and rule-based parsing for production pipeline throughput (4,700+ articles) and use LLMs only for sample evaluation.
Local Docker / Hyper-V driver conflicts (Medium)	Run Docker containers and Airflow services inside cloud-based GitHub Codespaces (Ubuntu VM) to avoid local host driver crashes.
Duplicate records on re-runs (High)	Enforce natural key uniqueness on `news_id` with `ON CONFLICT (news_id) DO NOTHING` in all database insert routines.
Database schema corruption or orphan rows (High)	Run automated pytest quality gates as the final DAG task; halt downstream alerts if any foreign key orphan is detected.

KPIs (Key Performance Indicators)

- **Pipeline Throughput & Speed:** Full batch processing of 4,700+ articles (Bronze to Silver) completes in under 3 minutes; daily incremental runs complete in under 30 seconds; database analytics queries run in under 200 ms.
- **System Uptime & Availability:** Streamlit dashboard uptime ≥ 99% during demonstration periods; Airflow schedule reliability ≥ 99% with automatic 2-retry policy and 5-minute backoff.
- **Data Quality & Integrity:** Exactly 0 orphan records between fact and dimension tables (100% referential integrity); 0% duplicate records; 100% pass rate on all 5 automated warehouse invariants.
- **Dashboard Usability:** Dashboard initial load under 5 seconds; filter and map re-rendering under 1 second.